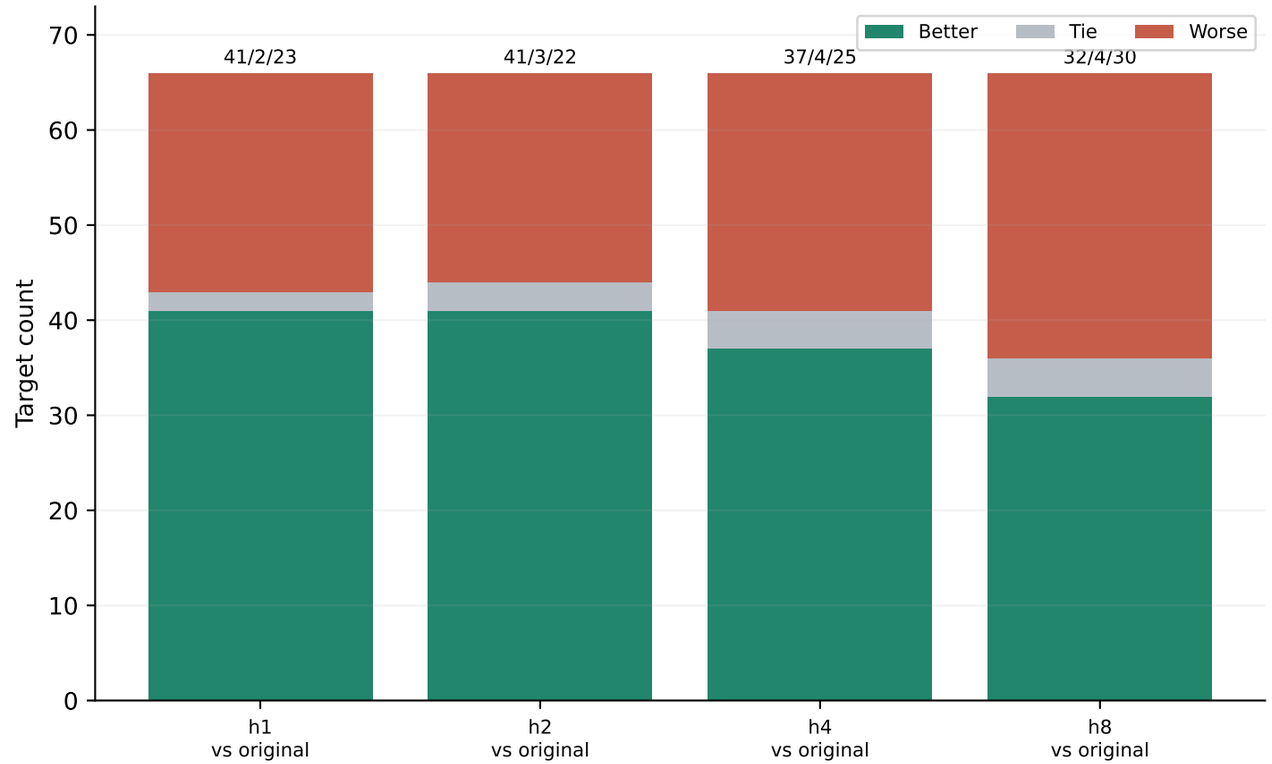
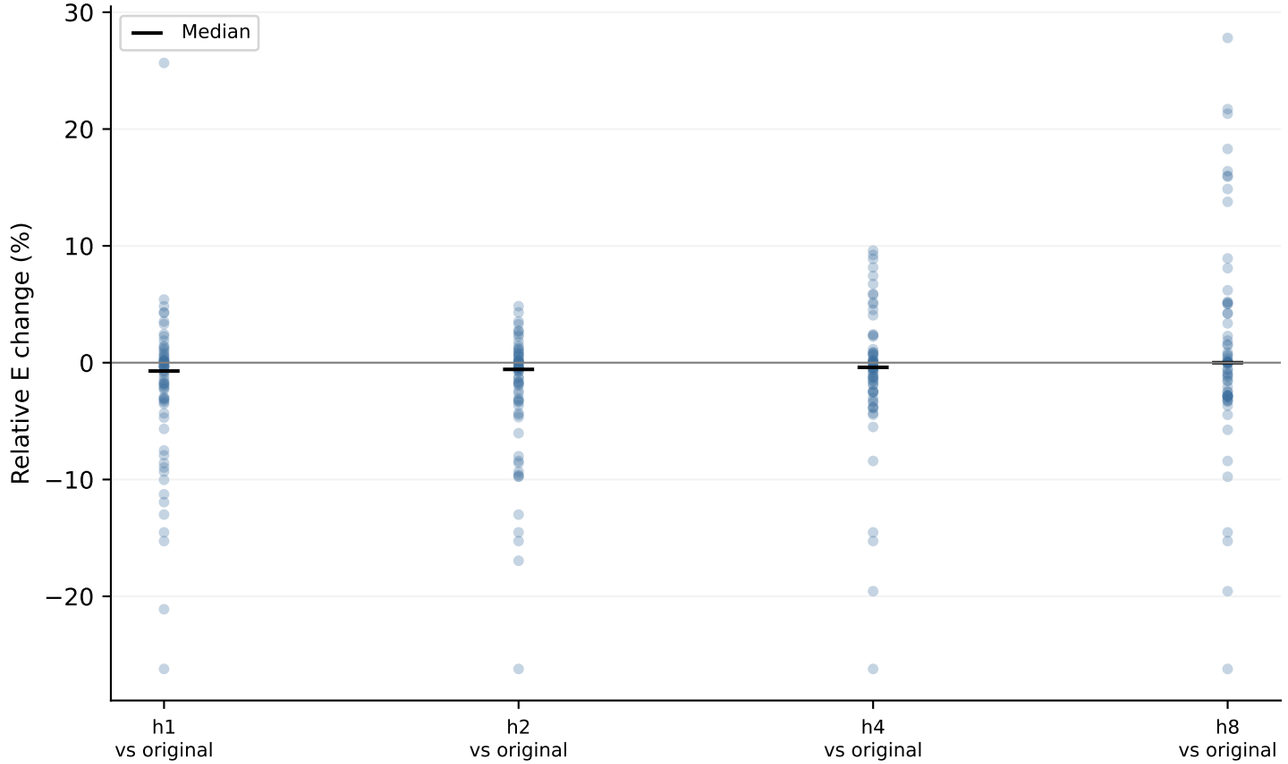
Phase 4: future-aware parking
Same Phase 3 no-cooling model; one measured timing sample per new variant

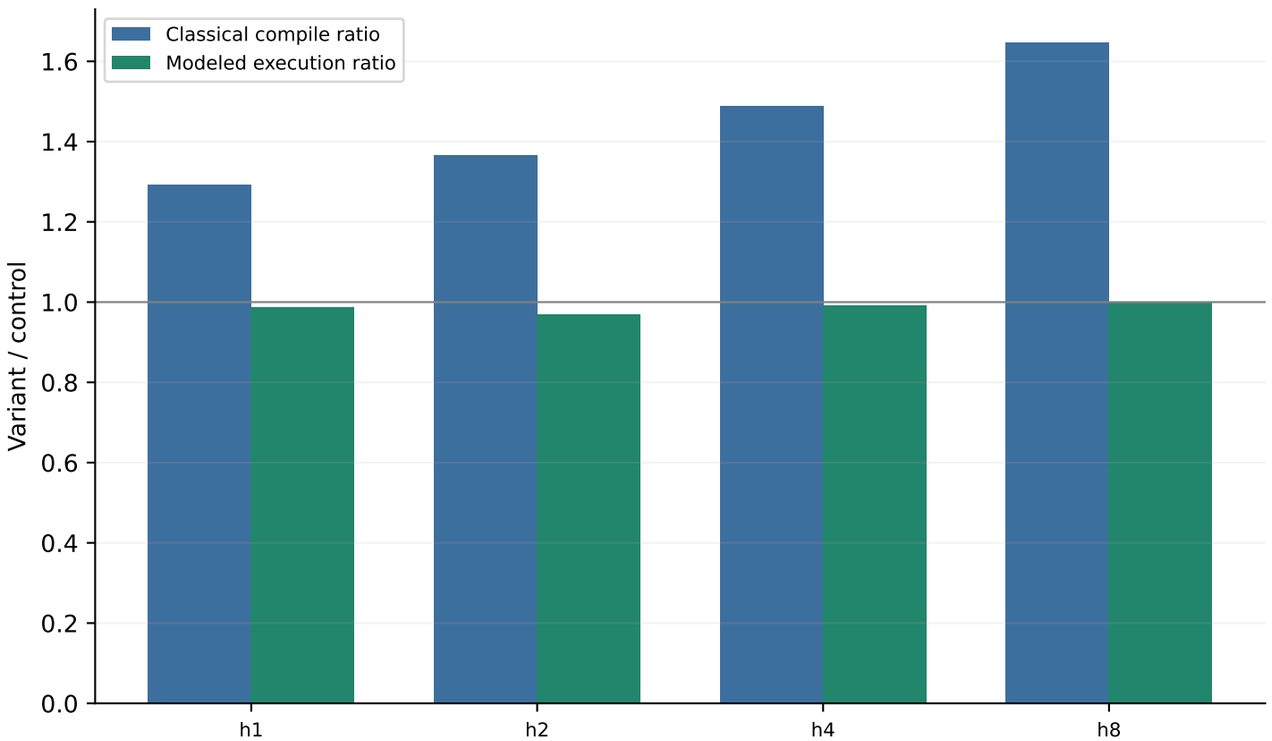
Paired outcomes (66 circuit/grid targets)



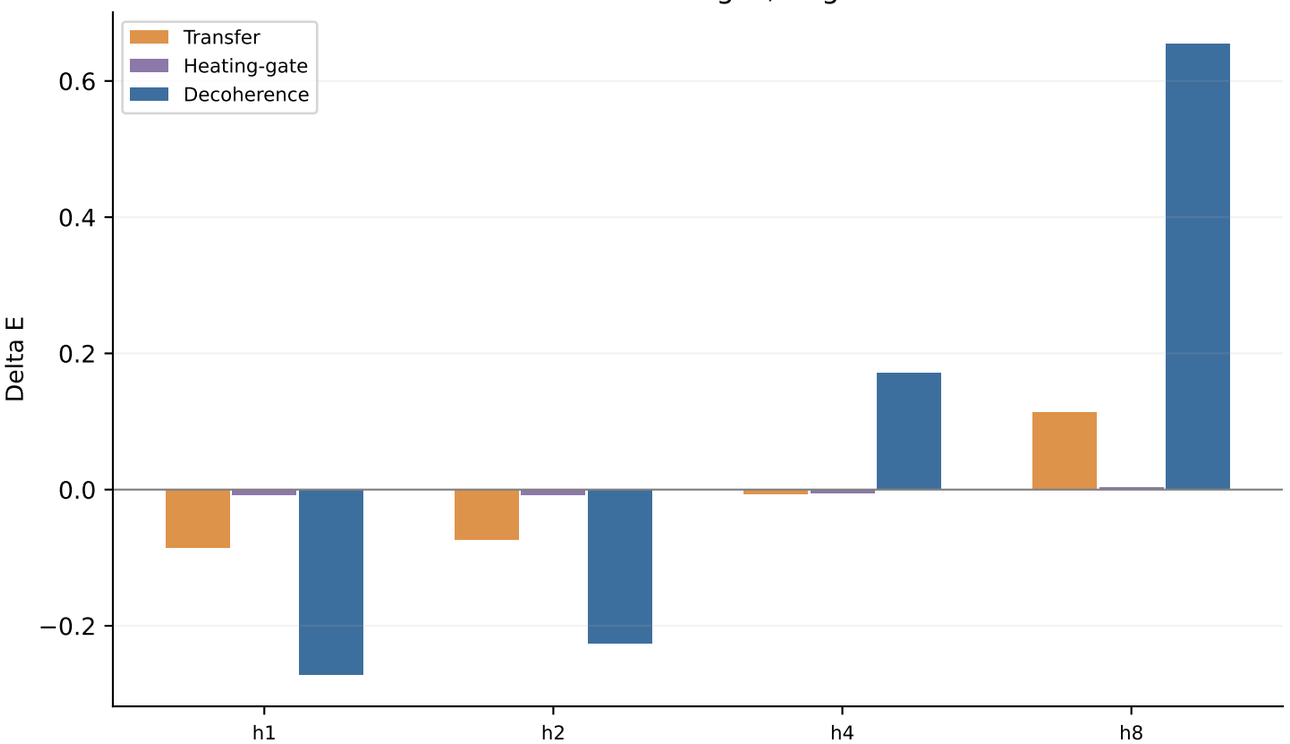
Error change distribution; negative is better



Paired median time ratios

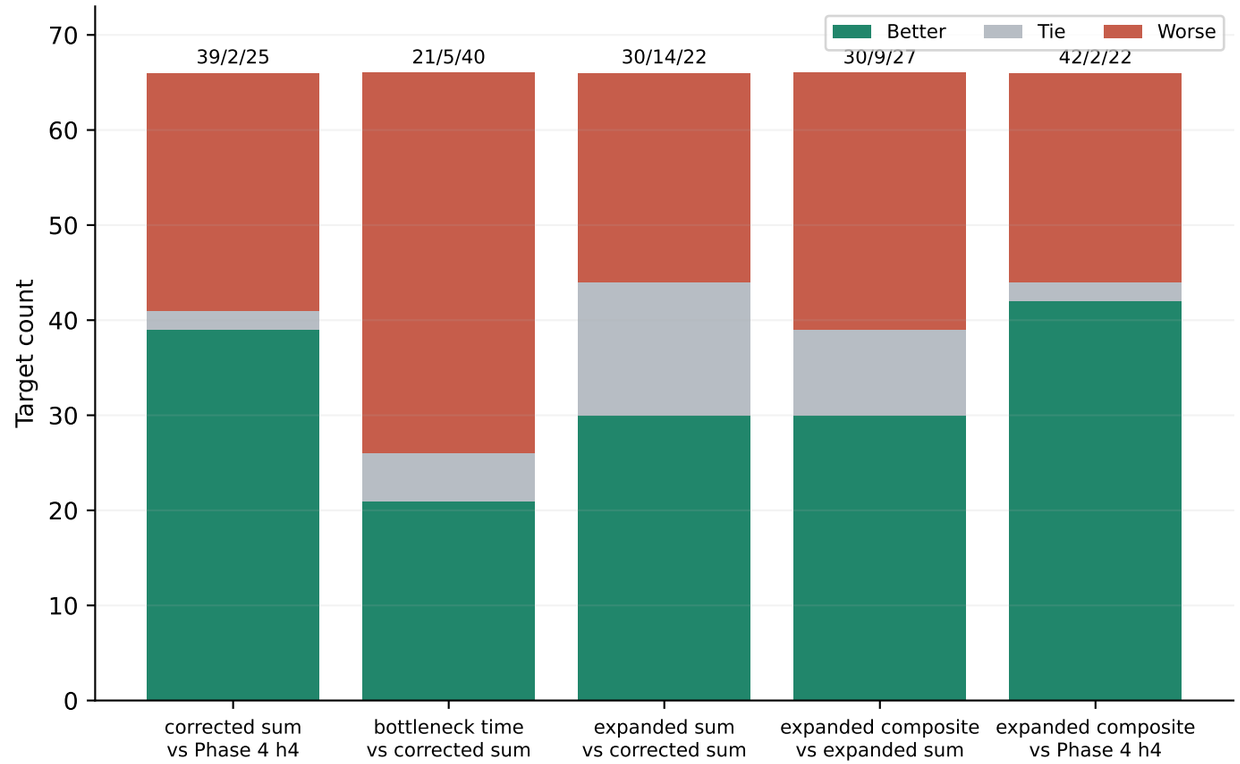


Mean additive error changes; negative is better

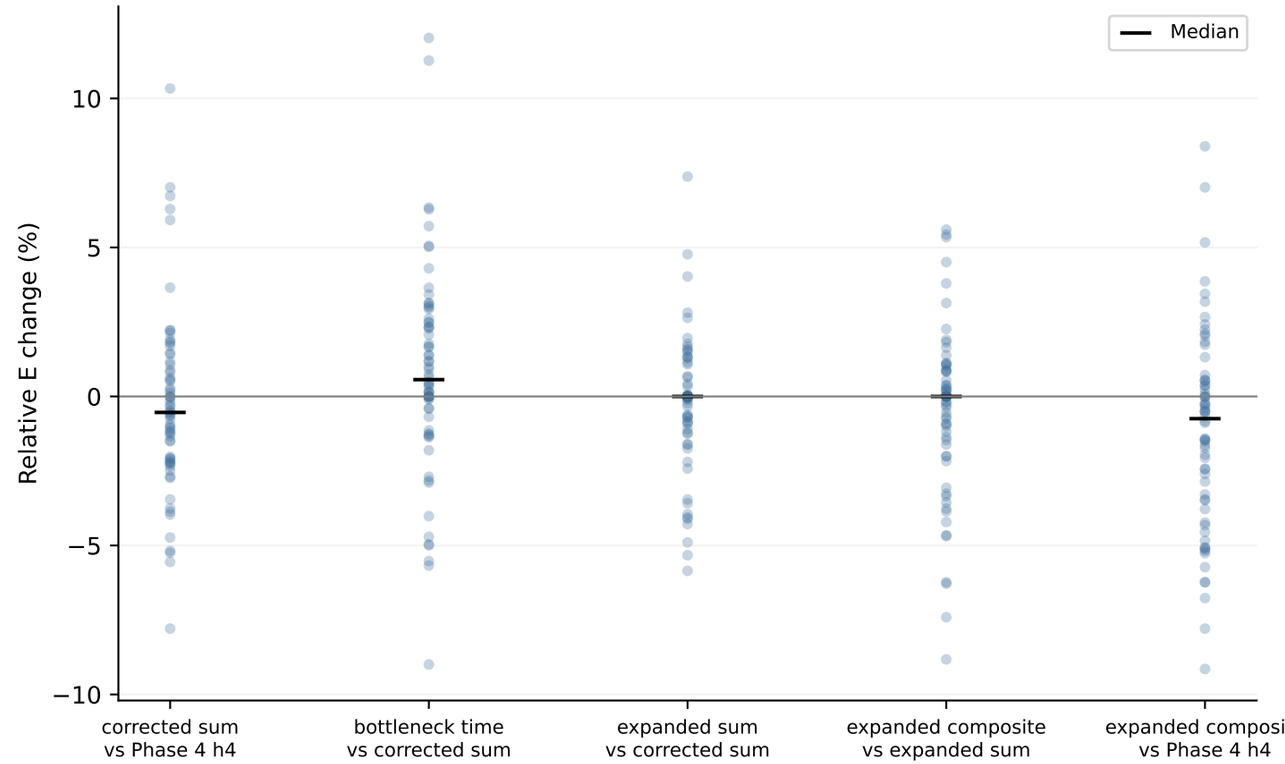


Phase 5: separate pipeline, candidate-budget and objective effects
Same Phase 3 no-cooling model; one measured timing sample per new variant

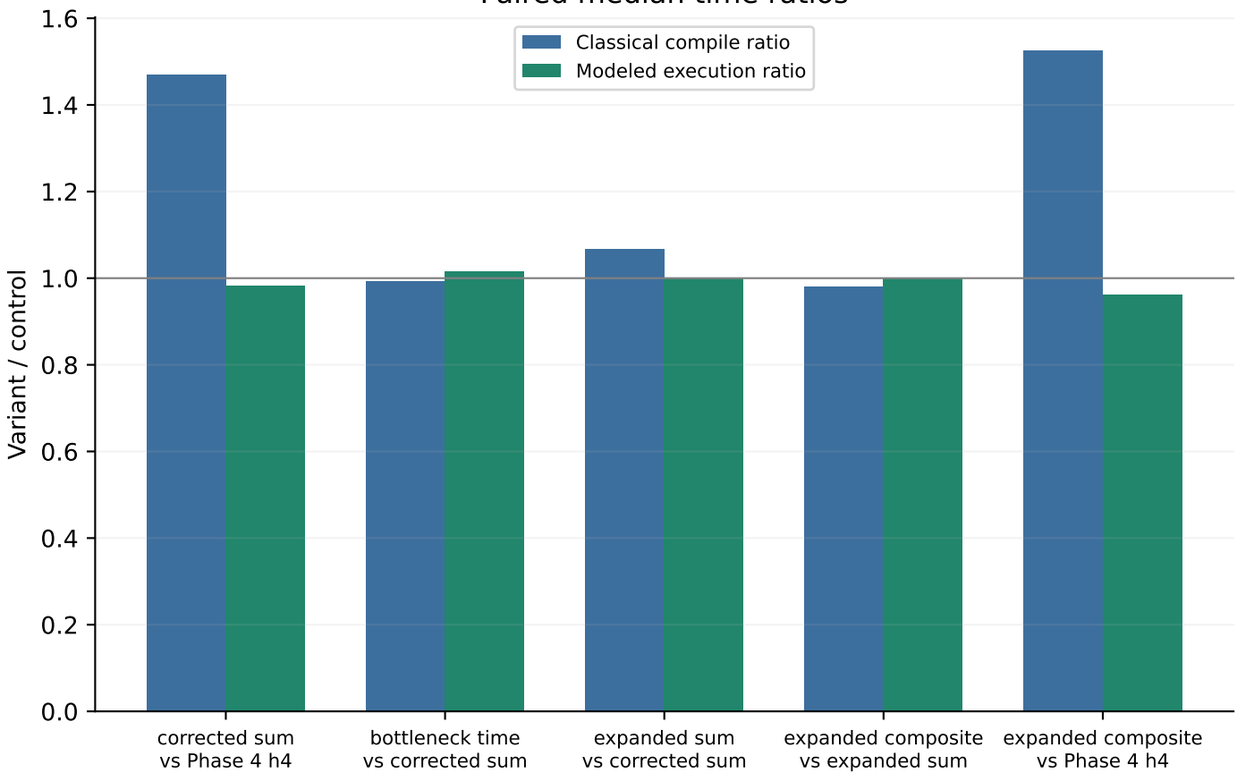
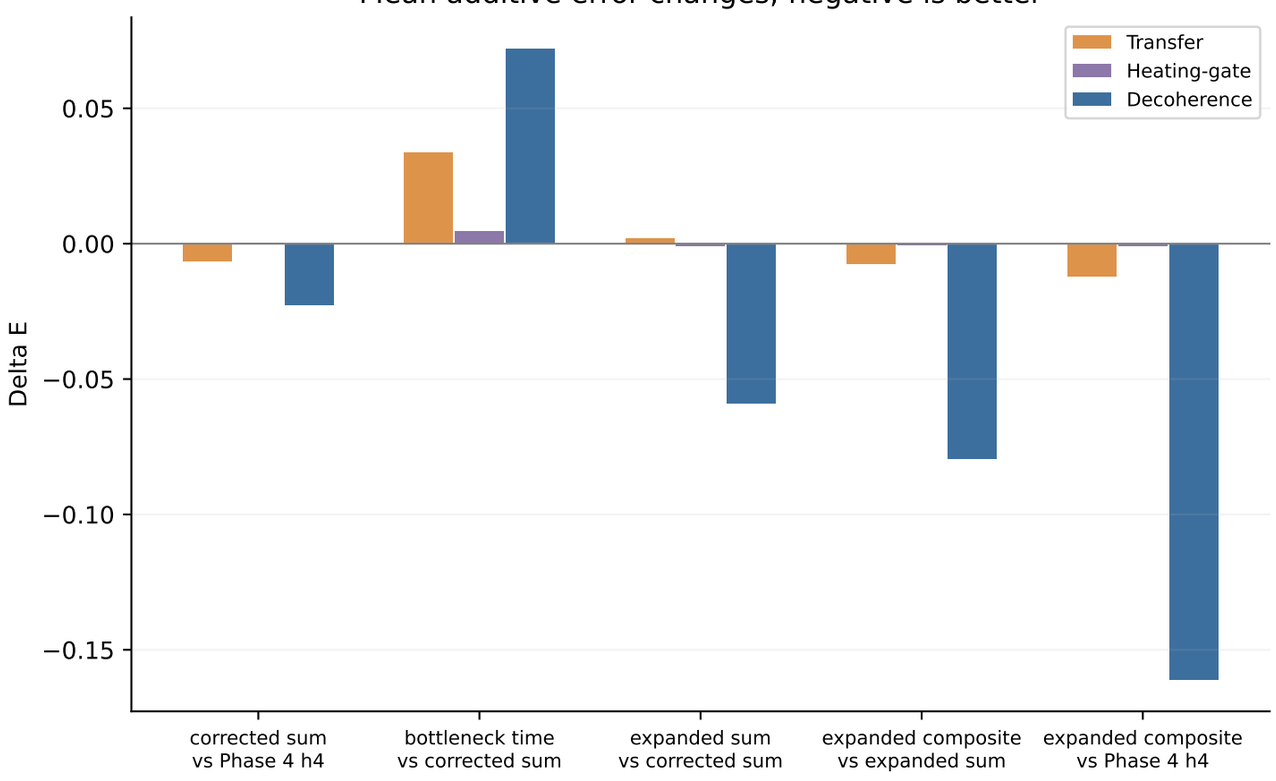
Paired outcomes (66 circuit/grid targets)



Error change distribution; negative is better



Paired median time ratios



Mean additive error changes; negative is better